

How teaching LLMs to reason step-by-step unlocks complex problem solving

Research Overview · 2025

1. Introduction

Large language models (LLMs) can answer many questions correctly when a direct answer is expected. But they struggle on tasks that require multi-step inference — arithmetic, symbolic reasoning, commonsense logic — when asked to respond with a single token or short phrase. **Chain-of-Thought (CoT) prompting** addresses this by encouraging the model to produce intermediate reasoning steps before arriving at a final answer.

The central insight, introduced by Wei et al. (2022), is deceptively simple: if a few-shot prompt includes example solutions that show their work, the model learns to show its work too — and in doing so, solves problems it previously could not.

Chain-of-Thought prompting is a technique in which a language model generates a sequence of intermediate reasoning steps that lead to a final answer, rather than the answer directly.

CoT prompting has become one of the most widely studied and applied techniques in modern NLP, leading to a family of related methods: Zero-Shot CoT, Self-Consistency, Tree-of-Thoughts, Least-to-Most prompting, and more. This document surveys the landscape.

2. Background & Motivation

2.1 The Failure Mode: Single-Step Answers

Standard few-shot prompting provides input–output pairs as demonstrations. The model is expected to pattern-match on these examples and produce a correct output. This works well for many tasks but fails when the reasoning path from question to answer is non-trivial.

Standard few-shot prompt (fails on hard math):

Q: Roger has 5 tennis balls. He buys 2 more cans of 3 balls each.
How many tennis balls does he have now?
A: 11

Q: The cafeteria had 23 apples. If they used 20 to make lunch and
bought 6 more, how many apples do they have?
A: ??? ← model outputs 27 (wrong)

2.2 The Fix: Reasoning Traces

Adding explicit reasoning steps to the demonstration examples dramatically improves accuracy, particularly on tasks that require multiple logical operations. The model is not being given the answer

— it is being shown how to think.

Chain-of-Thought few-shot prompt:

Q: Roger has 5 tennis balls. He buys 2 more cans of 3 balls each.

How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans $\times$ 3 balls = 6 balls.

$5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

A: They started with 23. Used 20: $23 - 20 = 3$.

Then bought 6: $3 + 6 = 9$. The answer is 9. ← now correct

2.3 Why Does It Work?

Several hypotheses have been proposed for why CoT improves accuracy:

- **Decomposition:** Breaking a hard problem into sub-steps reduces each step's difficulty.
- **Computation allocation:** Generating more tokens gives the model more 'thinking time.'
- **Error correction:** Intermediate steps expose incorrect assumptions mid-reasoning.
- **Training signal alignment:** Scratchpad-style outputs appear in pretraining data (math solutions, code comments, legal reasoning).

Empirical evidence shows that CoT benefits emerge strongly only above ~100B parameters in dense models. Smaller models benefit less, and can be harmed by being asked to generate reasoning chains they cannot.

3. Taxonomy of CoT Methods

3.1 Few-Shot CoT

The original formulation from Wei et al. (2022). A small number of (question, reasoning chain, answer) triples are placed in the prompt as demonstrations. The model is then given a new question and produces its own reasoning chain before the final answer.

Key design choices: quality of demonstrations, number of shots (typically 4–8), domain match between demonstrations and test questions, and whether the reasoning traces are hand-authored or model-generated.

3.2 Zero-Shot CoT

Kojima et al. (2022) showed that appending the phrase *"Let's think step by step."* to a question — with no examples — elicits a reasoning trace in sufficiently large models. Zero-Shot CoT requires no manual annotation of demonstrations and generalises remarkably well across task types.

Q: A juggler has 16 balls. Half of them are golf balls and half of those are blue. How many blue golf balls are there?

Let's think step by step.

- 16 balls total.
- Half are golf balls: $16 / 2 = 8$ golf balls.
- Half of those are blue: $8 / 2 = 4$.
- The answer is 4.

3.3 Self-Consistency

Wang et al. (2022) observed that a single greedy reasoning chain is brittle — a wrong intermediate step propagates to a wrong answer. Self-Consistency samples *k* diverse reasoning paths (via temperature sampling) and takes a majority vote over the final answers. This substantially improves accuracy on reasoning benchmarks at the cost of *k*× inference.

Method	Examples needed	Inference calls	Best for
Few-Shot CoT	4–8 annotated	1	Domain-specific, curated tasks
Zero-Shot CoT	None	1	General use, rapid prototyping
Self-Consistency	4–8 annotated	k (5–40)	High-stakes, math/science
Tree-of-Thoughts	Optional	Many	Search & planning problems
Least-to-Most	4–8 annotated	2+	Compositional generalisation

Table 1: Comparison of major Chain-of-Thought variants.

3.4 Tree of Thoughts (ToT)

Yao et al. (2023) extended CoT from a linear chain to a tree structure. The model generates multiple candidate next-thoughts at each step, evaluates them (either self-evaluated or via a separate scorer), and uses search algorithms (BFS/DFS) to navigate toward a solution. ToT excels on tasks with discrete search spaces such as the Game of 24 and crossword puzzles.

3.5 Least-to-Most Prompting

Zhou et al. (2022) proposed a two-stage approach: first decompose the problem into sub-problems, then solve them sequentially using the solutions of earlier sub-problems as context. This is especially

effective for compositional generalisation tasks where the test problem is harder than any seen during training.

4. Benchmark Results

CoT prompting has been evaluated on a wide range of reasoning benchmarks. The most commonly reported results cover mathematical word problems, commonsense reasoning, and symbolic manipulation.

4.1 Mathematical Reasoning

Benchmark	Standard	Few-Shot CoT	Self-Consist.	Model
GSM8K	17.9%	56.4%	74.4%	PaLM 540B
MATH	6.9%	33.0%	—	GPT-4
AQuA-RAT	46.0%	73.0%	79.0%	PaLM 540B
MultiArith	36.5%	93.0%	96.0%	PaLM 540B
SVAMP	69.9%	79.9%	86.8%	GPT-3 175B

Table 2: Accuracy on math reasoning benchmarks. Results from Wei et al. (2022) and Wang et al. (2022).

4.2 Commonsense and Symbolic Reasoning

Beyond mathematics, CoT improves performance on StrategyQA (+14 pp), CommonsenseQA (+8 pp), and BIG-Bench Hard tasks such as object counting, date arithmetic, and causal reasoning. Symbolic tasks including last-letter concatenation and coin-flip tracking show near-perfect accuracy with CoT but near-random performance without it.

On BIG-Bench Hard — a curated set of 23 tasks believed to be beyond the capabilities of current models as of early 2022 — CoT prompting with GPT-4 achieves above-human performance on many tasks.

5. Practical Design Principles

5.1 Writing Good Demonstrations

For few-shot CoT, the quality of reasoning traces matters more than their quantity. Empirical findings suggest:

- **Correctness:** Incorrect reasoning chains hurt accuracy even if the final answer is correct.
- **Clarity:** Explicit, step-labeled chains (Step 1: ... Step 2: ...) outperform fluent prose on structured tasks.

- **Diversity:** Demonstrations covering different problem types improve generalisation.
- **Length:** Longer chains are not always better; brevity reduces the chance of mid-chain errors.

5.2 Prompting Templates

Several phrasings consistently elicit reasoning chains:

Zero-Shot triggers:

- "Let's think step by step."
- "Let's work through this carefully."
- "Think through this problem step by step, then give your final answer."

Structured output template:

Reasoning: [model fills in steps]

Answer: [final answer]

Self-ask variant:

Are there any sub-questions I need to answer first?

Sub-question 1: ... Answer: ...

Sub-question 2: ... Answer: ...

Final answer: ...

5.3 When Not to Use CoT

CoT is not universally beneficial. Avoid it when:

- The task is simple classification or extraction where a short answer suffices.
- Latency is critical — generating a reasoning chain adds 2–5× tokens.
- The model is small (< ~7B dense parameters) and likely to produce unreliable chains.
- Verbosity could expose sensitive intermediate information.

6. Process Supervision & Alignment

A critical limitation of vanilla CoT is that it relies on outcome supervision — the model is rewarded only when the final answer is correct, even if the reasoning path was flawed or contained unjustified leaps.

Process Reward Models (PRMs) (Lightman et al., 2023) address this by providing step-level reward signals during RLHF.

OpenAI's "Let's Verify Step by Step" paper demonstrated that training a PRM on human-annotated step-correctness labels and using it to select the best of N sampled solutions (Best-of- N with PRM) substantially outperforms majority voting (Self-Consistency) at equal inference budget on MATH.

Process supervision matters for safety: a model that arrives at correct reasoning may generalise poorly to novel problems and may be encouraged to take steps that make a flawed intermediate step look plausible.

6.1 Reasoning Models

Models such as OpenAI o1/o3 and DeepSeek-R1 are trained end-to-end to produce long internal reasoning traces (sometimes called 'thinking tokens') before emitting a final answer. Unlike prompt-level CoT, these models have internalised the reasoning process through reinforcement learning, and their chain-of-thought is not directly shown to users in all interfaces. They represent the current frontier of CoT capability.

7. Limitations & Open Problems

7.1 Faithfulness

There is growing evidence that LLM reasoning chains are not always faithful — the model may produce a plausible-looking chain that does not causally reflect the process that produced the answer. Turpin et al. (2023) showed that inserting biasing hints changes both answers and reasoning traces without the chain acknowledging the bias.

7.2 Hallucinated Steps

Models can generate confidently-stated but factually incorrect intermediate claims. In multi-step chains, one hallucinated step can invalidate all subsequent steps. Retrieval-augmented CoT and tool-augmented CoT (e.g., code interpreters) mitigate this for knowledge-intensive and arithmetic tasks respectively.

7.3 Cost

CoT multiplies token usage — typically 3–10× for standard chains and 20–100× for reasoning models. For high-volume production workloads, the cost difference between direct and CoT inference is non-trivial and must be justified by accuracy requirements.

7.4 Prompt Sensitivity

Performance can vary significantly with minor changes to CoT trigger phrasing, demonstration ordering, and formatting. This brittleness makes evaluation and deployment harder than for standard prompting, and has motivated interest in automatic CoT generation (Auto-CoT, Zhang et al. 2022).

8. Key Papers & References

Wei et al. (2022)

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.
NeurIPS 2022. The foundational paper introducing few-shot CoT.

Kojima et al. (2022)

Large Language Models are Zero-Shot Reasoners.
NeurIPS 2022. Introduced the "Let's think step by step" zero-shot trigger.

Wang et al. (2022)

Self-Consistency Improves Chain of Thought Reasoning in Language Models.
ICLR 2023. Majority-vote over k sampled chains.

Yao et al. (2023)

Tree of Thoughts: Deliberate Problem Solving with Large Language Models.
NeurIPS 2023. Extends CoT to tree-structured search.

Zhou et al. (2022)

Least-to-Most Prompting Enables Complex Reasoning in Large Language Models.
ICLR 2023. Two-stage decomposition approach.

Lightman et al. (2023)

Let's Verify Step by Step.
ICLR 2024. Introduces process reward models for step-level supervision.

Turpin et al. (2023)

Language Models Don't Always Say What They Think.
NeurIPS 2023. Faithfulness analysis of CoT reasoning.

Zhang et al. (2022)

Automatic Chain of Thought Prompting in Large Language Models.
ICLR 2023. Auto-CoT: automated demonstration generation.

9. Summary

Chain-of-Thought prompting is one of the highest-leverage techniques available for improving LLM performance on reasoning tasks. By eliciting intermediate steps — either through few-shot demonstrations or a simple zero-shot trigger — it transforms problems that models previously failed at into problems they can solve reliably.

The method has evolved rapidly: from handcrafted few-shot chains to self-consistency voting, tree-structured search, process reward models, and fully trained reasoning models. Each advance trades inference cost for accuracy, and the practitioner's job is to find the right point on that curve for their use case.

Rule of thumb: if a problem has multiple steps, try Zero-Shot CoT. If accuracy is low, move to few-shot CoT with curated demonstrations. If accuracy is still low, try Self-Consistency. For maximum accuracy on hard reasoning tasks (o3, R1) with process supervision.